The Hidden Costs KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

1. Where does the answer come from? Draw the path from your Chromebook to the AI and back.

Chromebook → the school internet → a data center (computers, electricity, water) → back

2. The costs.

AI runs in buildings full of computers called data centers.

One text answer uses about 0.3 watt-hours of electricity and about 5 drops of water.

Workers in Kenya who labeled data for ChatGPT were paid less than $2 an hour.

3. Estimate the cost of our class.

Questions each student sends in one day (my guess)	20
Students in our class	18
Questions for the whole class (×)	360
Watt-hours (× 0.3)	108
Phone charges (÷ 15)	about 7
What the calculator said	108 Wh, 7.2 phone charges
The world, one day: U.S. homes for a day	25,000

One answer uses a little. Billions of answers use a lot: as much as thousands of homes.

4. Talk about it. AI is (worth / not worth) the cost when Answers vary.

The workers who label data should Answers vary. Be paid fairly and get support.

Go Further (Day 2). A receipt for AI. What did one AI answer or picture need?

RECEIPT

Item	Who pays?
electricity	company
water for cooling	people nearby
computer chips	company
people who labeled data	the workers
people who wrote the training text	writers, not asked

Which costs are paid by someone other than me? Answers vary.